Physik II — FS 2024 — Prof. R. Wallny — Serie 8

Release: 16th April 2024 – **Due date and discussion:** 25th April 2024

Exercise 1: Muons

Objective: time dilation, length contraction

The mean lifetime of a muon is 2.2 μs in the referendce frame of the muon. Let us consider that at a certain altitude h in the statosphere muons are created in a muon shower and radiated towards the earth with a velocity of $v = 0.99c$. Neglect the secondary interactions of the muons with the atmosphere. The number of muons decays exponentially with time $n(t) = n_0\, e^{-\alpha\, t}$ where α is a decay constant. The decay constant is the inverse of the lifetime $\alpha = 1/\tau$.

a) It is observed that on average 1% of the originally created muons arrive on the the earth. At which altitude have the muons been created?

b) How large is this distance in the reference frame of the muon?

c) What would be the probability for a muon to arrive on the earth if there was no relativistic time dilation?

Exercise 2: Non-commutativity of Lorentz transformations

Objective: Notation of transformations in matrix form. Differences of translations in non-relativistic three-dimensional space, and in the relativistic four-dimensional Minkowski space. Commutativity.

Lorentz transformations are coordinate transformations of the four-dimensional Minkowski space. Those transformations are not necessarily commutative (interchangeability of the order of application). We are going to verify this statement.

a) Consider two reference system R and R'. R' is moving with respect to R with constant velocity: $\vec{v}_1 = v \cdot \hat{x}$. The movement consists of pure translation. There is no rotation. The axis of the coordinate systems R and R', therefore, stay parallel to each other. Write down the matrix notation of the Lorentz transformation from R to R'. We will note $\Lambda(\vec{v}_1)$ the change of basis from R to R'.
Hint: We are in the relativistic regime, so v is close to c.

b) Consider a further reference system R" which is moving with constant velocity $\vec{v}_2 = v \cdot \hat{y}$ with respect to R'. Write down the Lorentz-transformation from R' to R" in matrix notation. $\Lambda(\vec{v}_2)$ is the notation for the change of basis.

c) Show that a transformation composed of the two previous transformations depends on the order in which we perform the transformations. In other words, show that the transformations are not commutative.

d) Discuss how a uniform translation of reference systems, which in the non-relativistic three dimensional case that we experience in every-day life is certainly commutative, can be non-commutative in Minkowski space? Give a simple analogue example of transformations in the three dimensional non-relativistic space, that are not commutative.

e) Now consider that the speed v is much smaller than c (non-relativistic case). Write down the transformations obtained above and check their commutativity.
Hint: Terms with order greater than 2 can be neglected in the Taylor development of γ.

Exercise 3: Time dilation

Objective: time dilation calculation and understanding.

A spaceship is flying away from Earth with a velocity $v = 0.866\,c$. It sends out two light signals, the second one $\Delta T' = 4\,\mathrm{s}$ after the first one (as measured in the reference frame of the spaceship).

a) Show that the time difference ΔT measured on earth between the arrival of the two signals (in the reference frame of Earth) equals

$$\Delta T = \sqrt{\frac{1 + v/c}{1 - v/c}}\Delta T' \tag{1}$$

b) What distance (in the reference frame of Earth) does the spaceship cover between the emissions of the two signals?

c) A body with a mass $m = 1$ kg is resting in the reference frame of the spaceship. What is its kinetic energy in the reference frame of Earth?

Exercise 4: Twin paradox using Doppler effect

Objective: relativistic Doppler effect, space contraction.

A space ship with brother B' departs from Earth with a velocity v close to the speed of light c towards the star α-Centauri. His twin brother A stays on Earth. The distance from Earth to α-Centauri as measured from Earth is L_0. When astronaut B' arrives at α-Centauri, he turns around and then flies back with the same velocity to the Earth.
The two brothers communicate by radio signals. Suppose they arrange to send each other one signal a year. Then each will know the other's age by just counting the total number of signals received between the moments of departure and return of the ship. Assume that the time needed for the space ship at α-Centauri to turn around is zero.

a) How many signals did each brother receive during the trip? Consider the problem first in the reference frame of Earth and then again in the reference frame of the space ship.

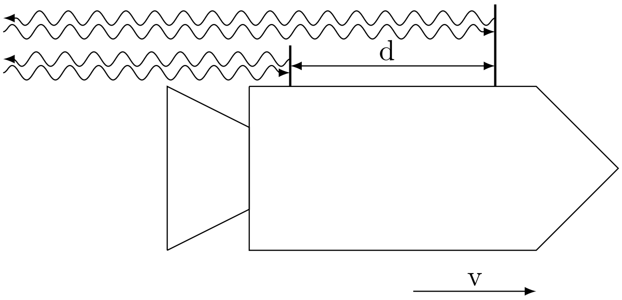
b) Which brother is younger after the trip is finished?

Hint: Consider the Doppler effect of light: if the space ship emits a signal with frequency f_0, then a different frequency f_1 will be measured in the reference frame of Earth, given by

$$f_1 = \sqrt{\frac{1 - \beta}{1 + \beta}}$$

Exercise 5: Velocity measurement using radar signals

Objective: Length contraction, invariance of the speed of light.



A rocket is moving away from the earth with velocity v. A radar signal emitted from a terrestrial station is reflected by two mirrors mounted on the rocket. One mirror is placed on the frontside, the other on the backside of the rocket. The distance between the two mirrors in the rest frame of the rocket is d. The terrestrial station detects the back-reflected radar signals separated in time by Δt. Determine the velocity v of the rocket as a function of Δt and d.